

Instructions for Students:

Compile all the following screenshots into a **single PDF document**. Every screenshot **MUST** be a full-screen capture showing your host computer's taskbar and current time. Cropped screenshots will receive zero credit. Where noted, you must run the date command immediately before taking the screenshot to timestamp your terminal.

Deliverable 1: The Hypervisor Topology

- **What to do:** Open the VirtualBox Manager. Select your pfSense VM and click "Settings" -> "Network".
- **Screenshot:** Capture the screen showing Adapter 2 or Adapter 3 settings.
- **Instructor Grading Key:** Look for "Attached to: Internal Network" and Promiscuous Mode explicitly set to "Allow All". If this is wrong, their lab didn't work.

Deliverable 2: Ubuntu Server IP & Netplan Verification

- **Command:** Run date and then run ip a on the Ubuntu Server.
- **Screenshot:** The terminal output.
- **Instructor Grading Key:** The student's terminal must display the interface enp0s3 successfully holding the IP address 192.168.2.100 handed out by the pfSense DHCP server.

Deliverable 3: Suricata Custom Rule Injection

- **Command:** Run date and then run cat /etc/suricata/rules/local.rules
- **Screenshot:** The terminal output showing the contents of the file.
- **Instructor Grading Key:** Verify the exact syntax of the rule is present: alert icmp any any -> any any (msg:"PING DETECTED - ALARM ACTIVE!"; sid:1000002; rev:1;).

Deliverable 4: pfSense Firewall Rules (WebGUI)

- **What to do:** From the Windows 11 VM, log into pfSense (https://192.168.1.1). Navigate to Firewall -> Rules.
- **Screenshot 4A:** The **OPT1 (DMZ)** rules tab.
 - *Instructor Grading Key:* Must show a rule allowing "Pass All Traffic" (Protocol: Any, Source: Any, Destination: Any).
- **Screenshot 4B:** The **LAN** rules tab.

- *Instructor Grading Key:* Must show the specific rule added during Scenario B allowing traffic to the 192.168.2.100 destination.

Deliverable 5: Scenario A (The Failed External Attack)

- **Command (Kali):** Ensure Kali is on the NAT network. Run `date` then `nmap -sV 192.168.2.100`.
- **Screenshot:** Capture both the Kali terminal and the Ubuntu terminal side-by-side.
- **Instructor Grading Key:** The Kali output *must* say the port is filtered. The Ubuntu terminal running `sudo tail -f /var/log/suricata/fast.log` must be blank/show no alerts because pfSense dropped the packets at the border.

Deliverable 6: Scenario B (The Successful Insider Attack)

- **Command (Kali):** Ensure Kali is on the Internal Network (LAN_Zone). Run `date`, then `sudo dhclient`, then launch the attack: `ping 192.168.2.100`.
- **Screenshot:** Capture both the Kali terminal and the Ubuntu terminal side-by-side.
- **Instructor Grading Key:** The Kali terminal must show successful ping replies. The Ubuntu terminal *must* show the log flooding with the exact alert text: "PING DETECTED - ALARM ACTIVE!".